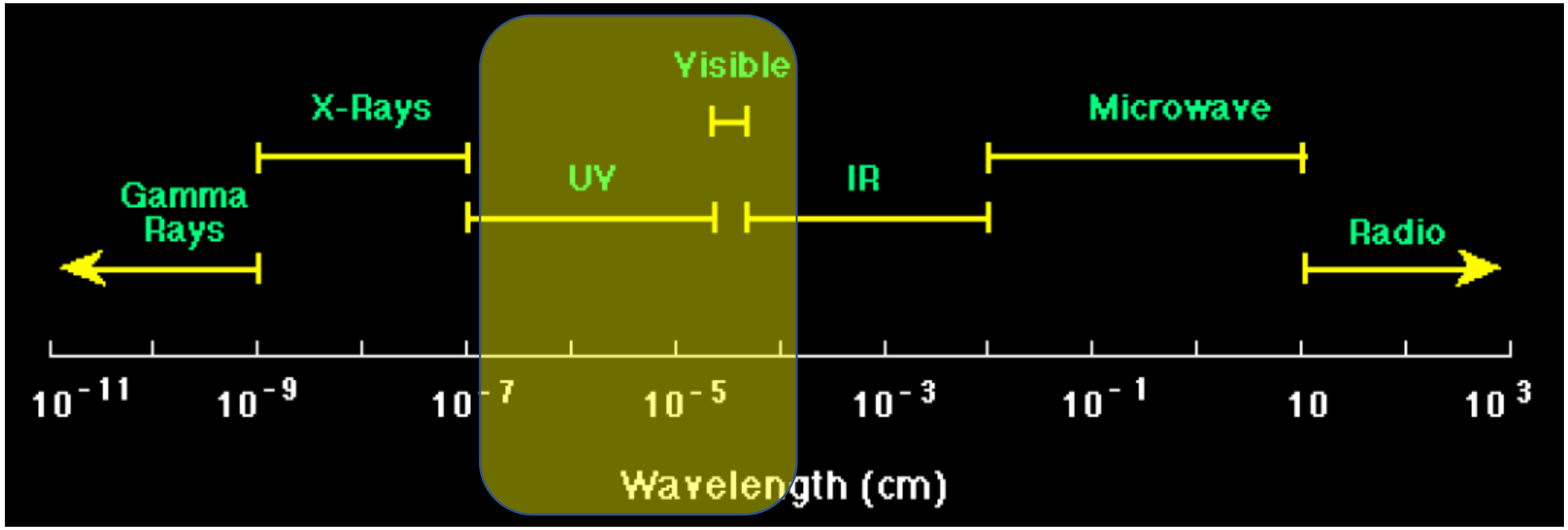
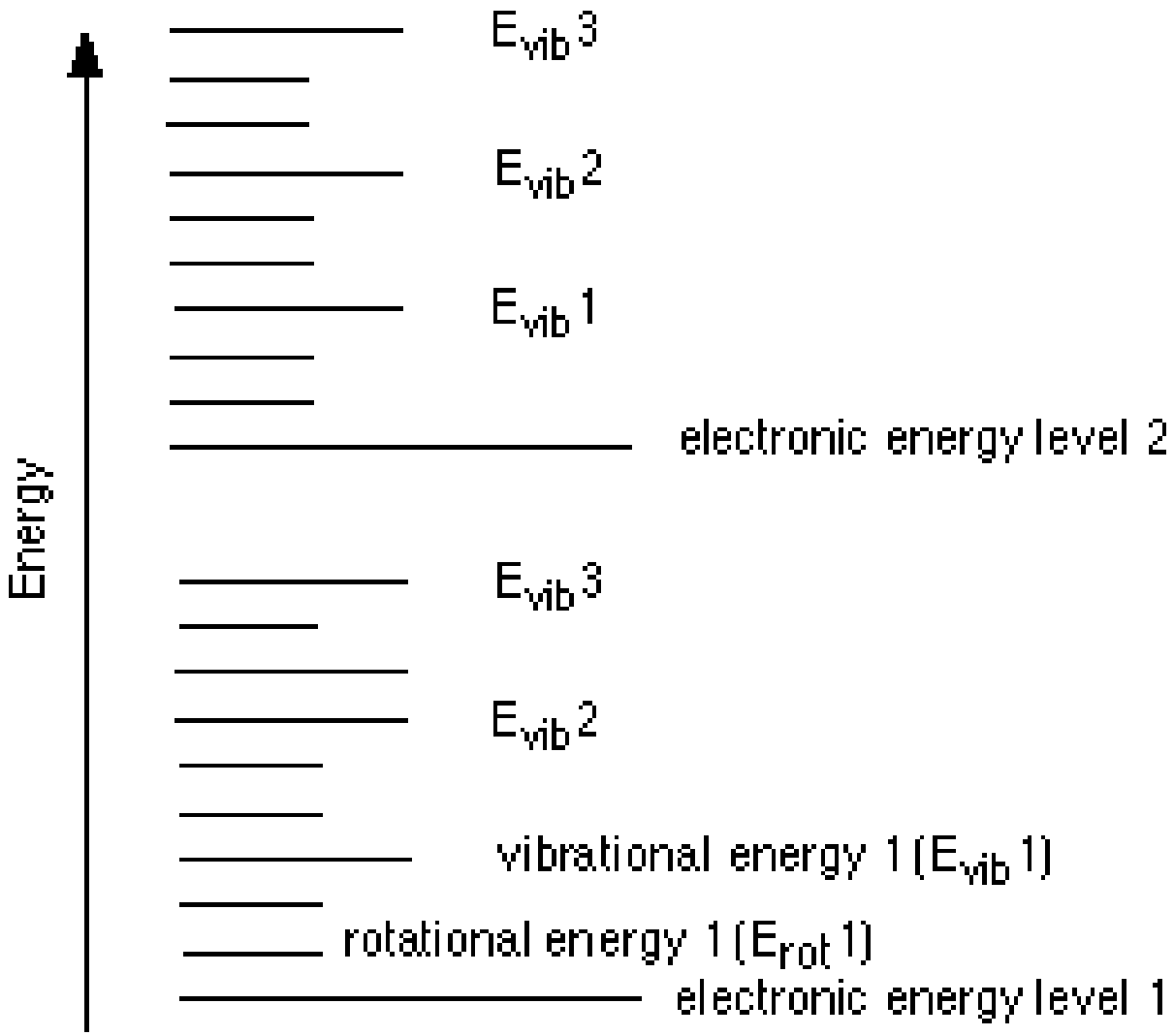
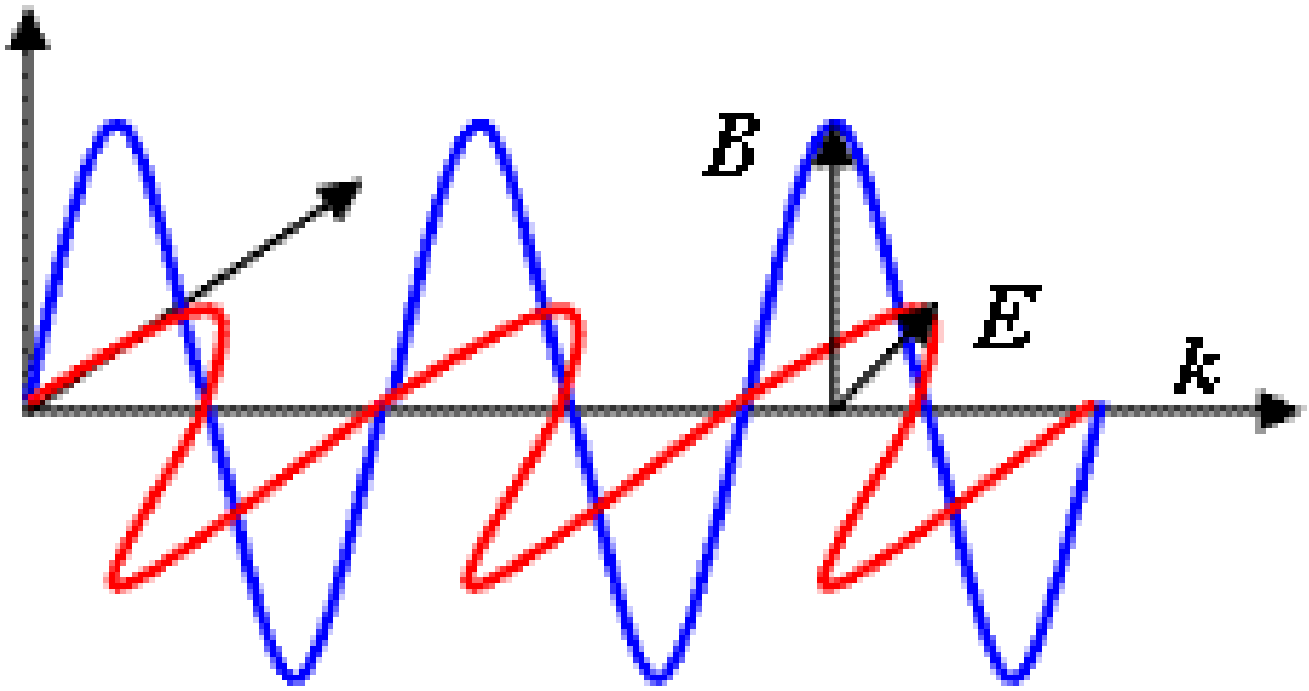
The regions of electromagnetic spectrum



Electronic Spectroscopy



Wave nature of radiation



Maxwell's Theory of Electromagnetic Radiation

Light travels in space in the form of **oscillating electric field**.

The **oscillating electric field** is generated by the acceleration and deceleration of charged particles which act as the source of radiation.

- If the particle moves with a <u>steady speed</u>
  $\rightarrow$ the field due to charge will follow the motion and the medium will be undisturbed.

- If there is <u>acceleration</u>
  $\rightarrow$ the field will not be able to follow the change.
  $\rightarrow$ a <u>disturbance</u> will be generated and <u>propagated</u> in space

The variation of the field strength as a function of time and space is given by:

$$E_x = E_0 \sin 2\pi \left(\nu t - \frac{x}{\lambda}\right)$$

E_x is the electric field strength vector in x-direction.
E_0 is a vector constant in time and space.

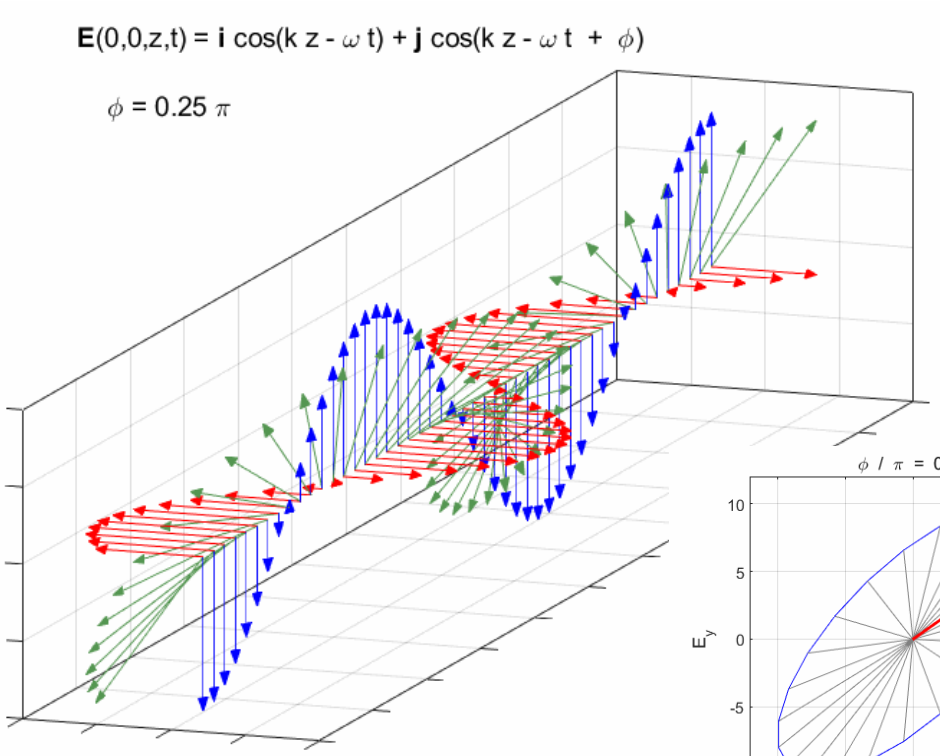
The electric vector is directed along the displacement direction of the wave $\rightarrow$ **direction of polarization**.

The plane containing the displacement vector$\rightarrow$ **plane of polarization**.

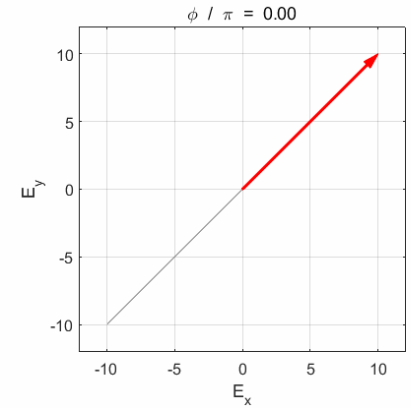
A **plane polarized** radiation oscillates only in one plane.

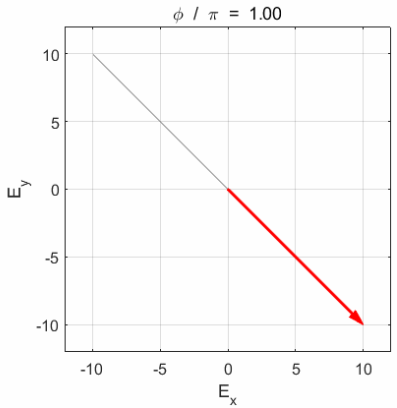
Using suitable optical devices radiation polarization in any direction can be obtained.

A combination of two wave trains polarized in different planes gives rise to **linearly**, **elliptically** or **circularly** polarized radiations.

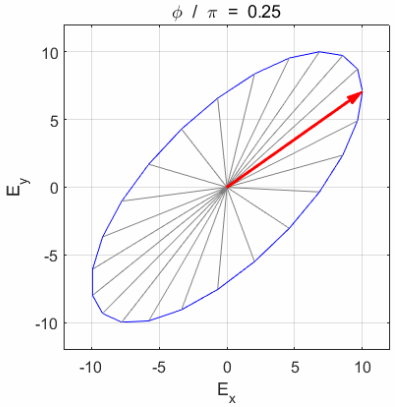


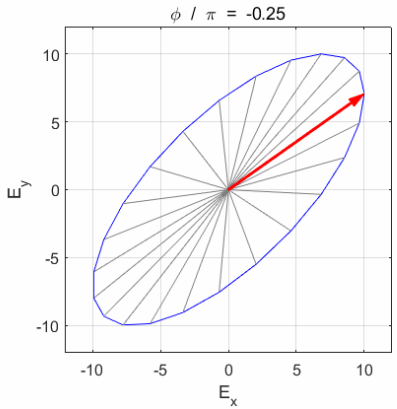
Vertically
Horizontally
Right Circularly
Polarized Light

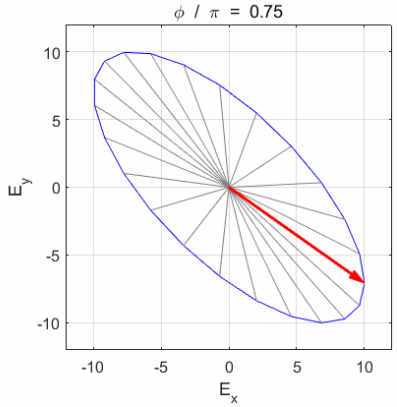


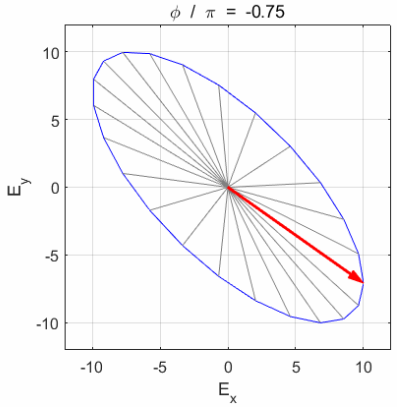


Linearly Polarized

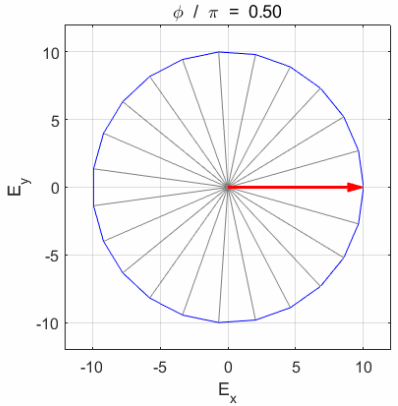


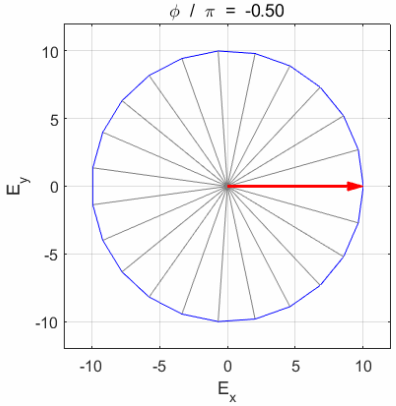




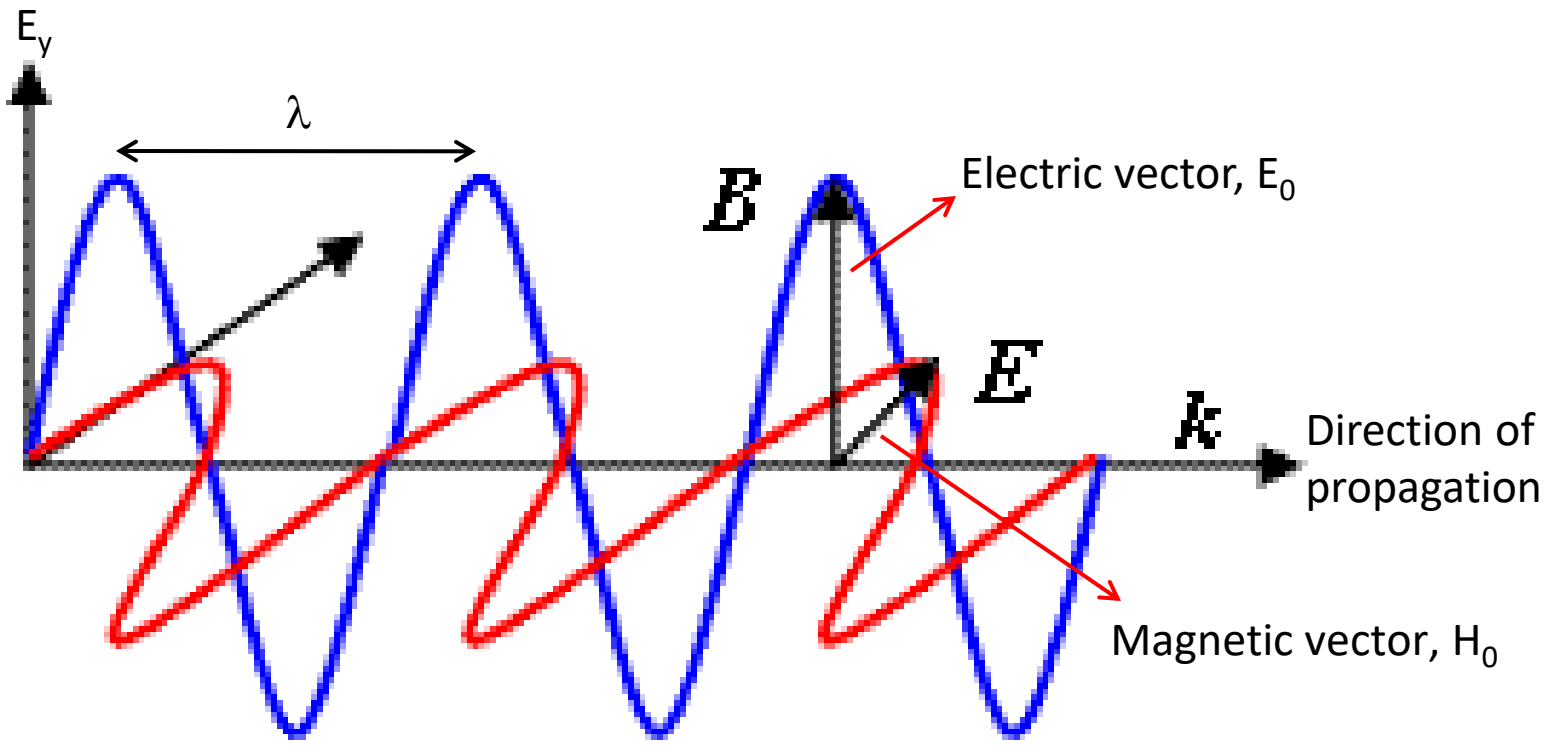


Elliptically Polarized





Circularly Polarized



λ = wavelength in nanometres.
ν = frequency or number of oscillations per second in Hertz (Hz).
A = amplitude of oscillation at any point *x.*
$c = \lambda\nu$ = velocity in ms^{-1}.
T = time period in s = 1/ν.
$\bar{\nu}$ = wave number in cm^{-1} = 1/λ.
ϕ = phase = x/λ.
$2\pi\phi = 2\pi x/\lambda$ = phase angle or the angle whose sine gives the displacement at any point.
I = intensity at any point x = square of displacement at that point.
node = point where the amplitude is zero.

$$E_x = E_0 \sin 2\pi\left(\nu t - \frac{x}{\lambda}\right)$$

Particle nature of radiation

A particle is defined by the mass *m* and its momentum *p* or energy *E*.

The particle nature of light is visualized in the form of wave packet or a quantum of radiation whose energy is given by $E = h\nu$.

The energy of photon is also given by Einstein's eqn. $E = mc^2$.

So, $mc^2 = h\nu$ Or, $mc = h\nu/c$ = momentum of a photon.

Dual nature of matter

In 1924, de Broglie emphasized the dual nature of matter.

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m(E-V)}}$$

E is the total energy and V is the potential energy.

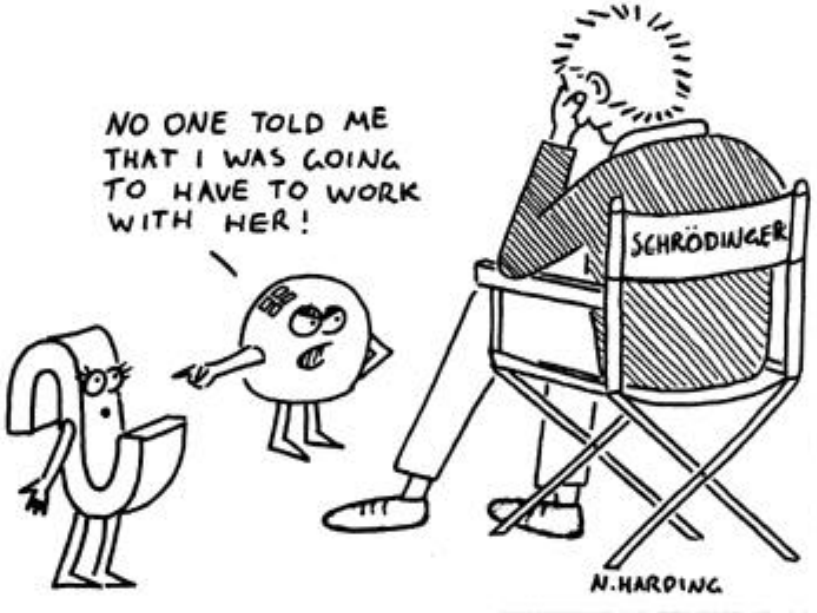
$$E = K.E. + P.E. = \frac{1}{2}mv^2 + V$$

$$E - V = \frac{1}{2}mv^2$$

$$m(E-V) = \frac{1}{2}m^2v^2 = \frac{1}{2}p^2$$

$$p^2 = 2m(E-V)$$

$$p = \sqrt{2m(E-V)}$$



The Schrödinger Equation

An expression for describing such a wave motion was obtained by **Schrödinger** in 1925.

The Schrödinger Equation is a second order differential equation which can be solved to obtain the total energy E of a dynamic system when expressed as a sum of kinetic and potential energies.

$$-\frac{h^2}{8\pi^2 m}\left[\frac{d^2}{dx^2}+\frac{d^2}{dy^2}+\frac{d^2}{dz^2}\right]\Psi + V\Psi = E\Psi$$

$$\left[-\frac{h^2}{8\pi^2 m}\nabla^2 + \hat{V}\right]\Psi = E\Psi$$

$$\mathrm{H}\Psi = E\Psi$$

$$\nabla^2 \equiv \frac{d^2}{dx^2}+\frac{d^2}{dy^2}+\frac{d^2}{dz^2} = \text{Laplacian operator}$$

$$\hat{V} \equiv V_x + V_y + V_z = \text{Potential energy operator}$$

$$H \equiv -\frac{h^2}{8\pi^2 m}\nabla^2 + \hat{V} = \text{Hamiltonian operator}$$

$$H\Psi = E\Psi$$

Ψ is known as the *eigenfunction* and E the *eigenvalue* or the characteristic energy value corresponding to this function.

On solving this equation, a number of values for Ψ such as $\Psi_0, \Psi_1, \Psi_2, \ldots, \Psi_n$ are obtained.

Corresponding to each eigenfunction, there are the characteristic energies $E_0, E_1, E_2, \ldots, E_n$.

Thus an electron constrained to move in a potential field can have only definite energy values for its motion.

Electronic energy states of atoms

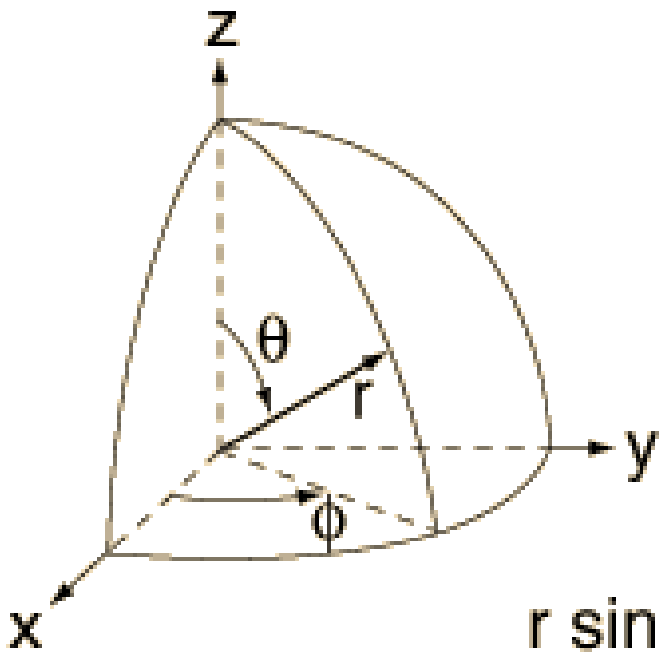
The hydrogen atom has a single electron confined to the neighbourhood of the nucleus by a potential field V, given by $-e^2/r$.

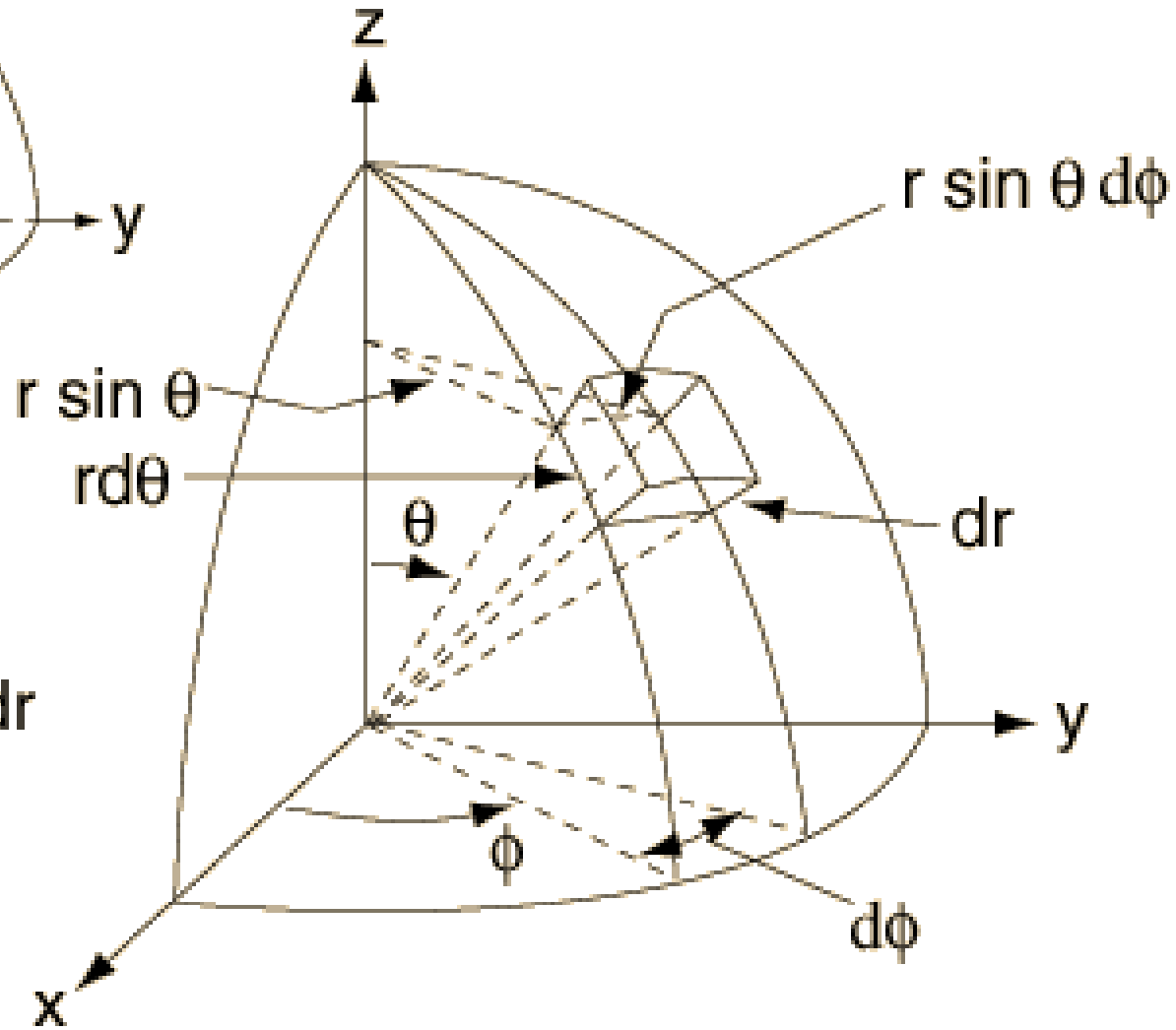
The solution of the appropriate Schrödinger equation is possible if the equation is expressed in polar coordinates r, θ and ϕ

$\rightarrow$ can be resolved into three independent equations each containing only one variable.

$$\Psi(r,\theta,\phi) = R(r)\Theta(\theta)\Phi(\phi)$$

Polar Coordinates





Volume element

$dV = r^2 \sin\theta \, d\theta \, d\phi \, dr$

The solutions of $R(r)\Theta(\theta)\Phi(\phi)$ equations introduce the three quantum numbers *n*, *l*, and *m* which have integral values.

n $\rightarrow$ the <u>principal quantum number</u> indicates the energy state of the system and is related to the dimension of the orbital.

l $\rightarrow$ the <u>azimuthal or orbital quantum number</u> originates due to the motion of the electron in its orbital.
This quantum number is related to the geometry or the shape of the orbitals and is denoted by the symbols s, p, d, f, etc. corresponding to the *l* values of 0, 1, 2, 3 respectively. There are *(n-1)* values for *l.*

m $\rightarrow$ the <u>magnetic quantum number</u> defines the orientation of the orbital in space.

m can have values 0, $\pm$1, $\pm$2, ……, $\pm l$.

s $\rightarrow$ the <u>spin quantum number</u> arises due to the spinning of an electron on an axis.

s can have two values +1/2 and -1/2.

Interaction of spin and orbital angular momenta

Orbital angular momentum is a vector quantity whose value is given by:

$$\sqrt{l(l+1)}\,\frac{h}{2\pi}$$

Spin angular momentum is a vector quantity whose value is given by:

$$\sqrt{s(s+1)}\,\frac{h}{2\pi}$$

The spin and orbital angular momenta of the electron are expected to interact with each other.

The resultant angular momentum can be predicted by the vector addition rule.

With only one electron in the unfilled energy shell, the orbital angular momentum and the spin angular momentum add up vectorially to give the resultant as:

$$\sqrt{j(j+1)}\,\frac{h}{2\pi}$$

Where *j* is the total angular momentum quantum number.

For one-electron system there can be only two values of *j* for a given *l* state:

$$j = l + \frac{1}{2} \text{ or } l - \frac{1}{2}$$

When the number of electrons is more than one, there are more than one possibilities for such interactions.

In a completed shell or a subshell the contributions of individual electrons cancel each other and the total angular momentum is zero.

For two electrons in an unfilled shell where the orbital angular momenta are denoted by l_1 and l_2 and the spin angular momenta by s_1 and s_2 , the possible interactions are:

(1) l_1 with l_2 to give L, and s_1 and s_2 to give S, followed by interaction between L and S to give J. $\rightarrow$ Russell-Saunders coupling.

(2) l_1 with s_1 to give j_1 and l_2 with s_2 to give j_2, followed by interaction between j_1 and j_2 to give J. $\rightarrow$ *j-j* coupling.

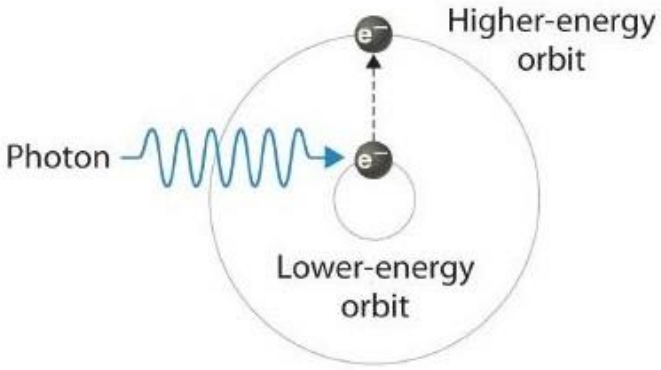
From the vector addition rule and the constraint that the values must differ by one in the unit of $h/2\pi$, the possible values of L and S are:

$$L = (l_1 + l_2), (l_1 + l_2 - 1), \ldots, |l_1 - l_2|$$
$$S = (s_1 + s_2), (s_1 + s_2 - 1), \ldots, |s_1 - s_2|$$

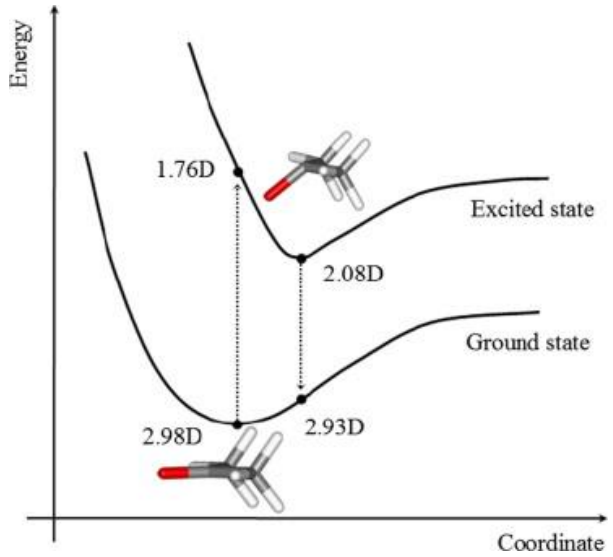
Therefore: $$J = L + S, L + S - 1, \ldots, |L - S|$$

For N number of electrons, the vector addition is carried out one by one to get the total L. Similarly, the total S is obtained. The two are finally coupled to get the total J.

Absorption of radiation causes the electrons to crossover from a lower energy electronic state to a higher energy electronic state.



As a result of this transition, the electronic configuration of the molecule changes and hence the excited state becomes polarized.



Accordingly, the electronic transition is accompanied by a **change of dipole moment** (also known as instantaneous dipole moment).

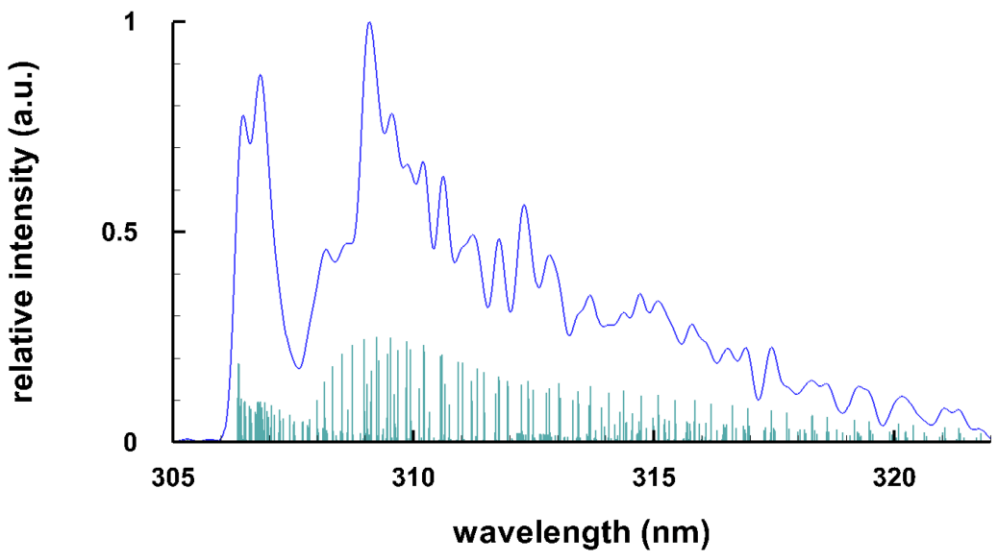
- The transition dipole moment integral is:

$$\langle \mu \rangle = \int \psi^* \hat{\mu} \psi d\tau$$

- where ψ* is the excited state wave function, $\hat{\mu}$ is the dipole moment operator, ψ is the ground state wave function, and dτ is the integration element.

Since each electronic state is associated with its vibrational and rotational states

→ the electronic spectra of heteronuclear and homonuclear diatomics in the vapour phase will exhibit **rotational and vibrational coarse structures**.



The **rotational fine structure** is not normally observed in UV-visible absorption spectrum.

→ because of small energy difference between the adjacent rotational energy states as compared to that of the vibrational energy states.

Raman spectrum mostly deals with the ground electronic state

→ **electronic spectrum** provides information also about the excited state.

Since the rotational constants of isotopes are different

→ they can be detected and their percentage abundance can be measured by comparing the intensity of the respective bands.

The vibrational parameters

→ (dissociation energy, frequency of the fundamental vibrational mode and hence the force constant) can be deduced for both the ground as well as the excited states from the vibrational coarse structure.